

Generation Interconnection System Impact Study Report

For Queue Project Q21-1931

Skimmer 138 kV

May 24, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1931. The Interconnection Customer proposes a 82.2 MW battery energy storage facility interconnecting at the Skimmer 138 kV substation in the Virginia area. The request was received on November 14, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 563 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,421,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Plaquemine Cogeneration Plant - Balcony Falls 138kV line	Reusens - Pentland 138kV	132.8	224	100.0
S Texa - Nacoochee 138kV line	Opossum Crk - Parkin 138kV	127.5	659	29.6
Butte - Balcony Falls 138kV	Reusens - Pentland 138kV L	121.9	234	100.0
Way Dm - Plaquemine Cogeneration Plant 138kV L	Reusens - Pentland 138kV	117.4	264	100.0
White Pine - Cusaw 138kV	Pentland - Resens 138kV	116.5	274	100.0
Way Dam Substation - Cushaw 138kV	Pentland - Reusens 138kV	116.3	276	100.0
White Pine - Devil'S Hill 138kV	Reusens - Pentland 138kV line	116.1	280	100.0
Snowden - Isla Grande Gis 11191120 138kV	Reusens - Pentland 138kV	115.7	285	100.0
Isla Grande Gis 11191120 - Devil'S Hill 138kV	Rusens - Pentland 138kV L	115.6	286	100.0
Walerga-Galbrath - Butte 138kV L	Reusens - Pentland Substation 138kV	111.5	245	100.0
La Joie Substation - Monel 138kV	Franklin-Elk Grove - New London 138kV	105.1	212	5.4
Pentland - Reusens 138kV	Devil'S Hill - Isla Grande Gis 11191120 138kV	101.6	325	100.0
Skimmer - Snowden 138kV	Reusens - Pentland 138kV	101.0	326	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Plaquemine Cogeneration Plant - Balcony Falls 138kV	Rebuild line with higher-capacity conductor	\$5,720,000	27% (\$1,558,000)
Nacoochee Substation - S Texa 138kV	Rebuild line with higher-capacity conductor	\$6,346,000	5% (\$290,000)
Butte - Balcony Falls 138kV L	Rebuild line with higher-capacity conductor	\$9,431,000	27% (\$2,569,000)
Way Dam - Plaquemine Cogeneration Plant 138kV	Rebuild line with higher-capacity conductor	\$2,649,000	27% (\$721,000)
White Pine - Cusaw 138kV	Rebuild line with higher-capacity conductor	\$3,435,000	27% (\$936,000)
Way Dam Substation - Cushaw 138kV	Rebuild line with higher-capacity conductor	\$4,049,000	27% (\$1,103,000)
White Pine - Devil'S Hill 138kV	Rebuild line with higher-capacity conductor	\$7,709,000	27% (\$2,100,000)
Isla Grande Gis 11191120 - Snowden 138kV	Rebuild line with higher-capacity conductor	\$2,996,000	27% (\$816,000)
Isla Grande Gis 11191120 - Devil'S Hill 138kV	Rebuild line with higher-capacity conductor	\$2,890,000	27% (\$787,000)
Butte Substation - Walerga-Galbrath 138kV	Rebuild line with higher-capacity conductor	\$13,298,000	27% (\$3,622,000)
La Joie Substation - Monel 138kV	Reconductor line	\$4,921,000	2% (\$121,000)
Reusens - Pentland 138kV	Reconductor line	\$2,134,000	27% (\$581,000)
Snowden - Skimmer 138kV	Reconductor line	\$796,000	27% (\$217,000)
	Total allocated to project		\$15,421,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.